

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data set A consists of **10** positive integers less than **60**. The list shown gives **9** of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these **9** integers is **42**. If the mean of data set A is an integer that is greater than **42**, what is the value of the largest integer from data set A?

Correct Answer: 52

Rationale

The correct answer is **52**. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of **10** values, **9** of which are shown. Let x represent the **10th** data value in data set A, which isn't shown. The mean of data set A can be found using the expression $\frac{43+45+44+43+38+39+40+46+40+x}{10}$, or $\frac{378+x}{10}$. It's given that the mean of the **9** values shown is **42** and that the mean of all **10** numbers is greater than **42**. Consequently, the **10th** data value, x , is larger than **42**. It's also given that the data values in data set A are positive integers less than **60**. Thus, $42 < x < 60$. Finally, it's given that the mean of data set A is an integer. This means that the sum of the **10** data values, $378 + x$, is divisible by **10**. Thus, $378 + x$ must have a ones digit of **0**. It follows that x must have a ones digit of **2**. Since $42 < x < 60$ and x has a ones digit of **2**, the only possible value of x is **52**. Since **52** is larger than any of the integers shown, the largest integer from data set A is **52**.